UNIT 18

INFORMATION & COMMUNICATION TECHNOLOGY (ICT)

INFORMATION TECHNOLOGY (IT)

Information Technology (IT) encompasses the utilization of computer systems, software, and networks to store, retrieve, transmit, and manipulate data for various purposes, such as processing information, automating tasks, and enabling communication.

COMMUNICATION TECHNOLOGY (CT)

Communication Technology (CT) refers to the tools, systems, and methodologies employed to facilitate the exchange of information and ideas among individuals or devices, encompassing mediums like telecommunication networks, internet protocols, and multimedia platforms.

COMPONENTS OF COMPUTER BASED INFORMATION SYSTEM (CBIS)

The five components of a computer-based information system are:

1. Hardware: This includes the physical equipment that makes up the system, such as computers, servers, storage devices, input/output devices (keyboard, mouse, monitor), and networking equipment.

2. Software: This refers to the programs and applications that enable the computer system to perform specific tasks. It includes operating systems, application software, and utility programs.

3. Data: Data is the raw information that the system processes. It can be in the form of text, numbers, images, videos, or any other type of digital content.

4. Procedures: These are the set of instructions or guidelines that dictate how the system should be used, how data is processed, and how tasks are performed. Procedures ensure that the system operates efficiently and effectively.

5. People: People are the users, operators, administrators, and stakeholders who interact with the system. They input data, use software applications, follow procedures, and make decisions based on the information generated by the system.

TRANSMISSION OF ELECTRICAL SIGNAL THROUGH WIRES

The process of transmitting electrical signals through wires involves the movement of electrons within the wire to convey information or power. Here's a simplified explanation of the process:

1. Source of Signal: The origin point generating an electrical signal, often voltage or current.

2. Signal Encoding: The process of representing information using variations in voltage or current.

3. Electron Movement: The motion of electrons within a wire due to an applied electric field.

4. Drift Velocity: The slow net speed at which electrons move in a wire due to their random motion and collisions.

5. Signal Propagation: The transmission of the electrical signal along the wire through a chain reaction of electron movements.

6. Signal Loss: The reduction in signal strength due to resistance and heat generation within the wire.

7. Interference and Noise: External factors causing distortion in the transmitted signal, including electromagnetic interference and noise.

8. Reception: The process of detecting, amplifying, and processing the transmitted signal for information recovery.

9. Output: Utilizing the processed signal for various purposes such as display, sound production, or control of devices.

TRANSMISSION OF RADIOWAVES THROUGH SPACES

The process of transmitting radio waves through space involves the propagation of electromagnetic waves, specifically radio frequency (RF) waves. Here's a simplified explanation of how this process works:

1. Signal Generation: Creation of alternating current at a specific frequency.
2. Modulation: Variation of radio wave properties to encode information.
3. Antenna Transmission: Conversion of electrical signal to electromagnetic wave by the antenna.
4. Emission into Space: Release of electromagnetic waves (radio waves) into the environment.
5. Propagation through Space: Travel of radio waves through the vacuum of space or atmosphere.
6. Line of Sight and Obstacles: Course alteration due to obstructions and Earth's curvature.
7. Reception: Capture of incoming radio waves by a receiving antenna.
8. Demodulation: Extraction of original information by reversing modulation.
9. Amplification and Processing: Strengthening and refining of the received signal.
10. Output: Utilization of processed signal for sound, display, or other applications.

FAX MACHINE

A fax machine transmits images as electrical signals over phone lines. It encodes images, sends them as audio tones, and at the other end, decodes and prints the received image.

CELL PHONE

A cell phone emits radio signals encoding voice and data, which travel to a tower. The tower relays these signals, allowing the receiving phone to decode and display them.

PHOTOPHONE

The photo phone conveys sound by changing light intensity with sound waves. The light carries the encoded sound, which is turned back into sound at the receiver.

TRANSMISSION OF LIGHT SIGNALS THROUGHOPTICAL FIBRES

Light signals are transmitted through optical fibers using a process called optical communication, which involves sending light pulses through thin strands of glass or plastic fibers. Here's how the process works:

1. Source of Light Signal: The origin point generating intense light pulses for transmission.
2. Signal Encoding: The process of modulating light pulses to encode information.
3. Transmission into Optical Fiber: Introduction of modulated light pulses into the core of an optical fiber.
4. Total Internal Reflection: Phenomenon where light repeatedly bounces within the core of the optical fiber due to refractive index differences.
5. Propagation through Fiber: Travel of light pulses through the fiber via total internal reflection.
6. Signal Attenuation and Dispersion: Loss of signal strength and slight separation of different wavelengths due to scattering and other factors.
7. Reception and Demodulation: Detection of light pulses at the receiving end and extraction of original information through demodulation.
8. Amplification and Processing: Strengthening of received signal and reduction of noise

9. Output: Utilization of the processed signal for various applications such as sound reproduction or data display.

INFORMATION STORAGE DEVICES

Information Storage Devices are tools or media used to store digital data for later retrieval and use. These devices hold information in various formats and capacities, enabling data to be preserved, shared, and accessed. Different types of storage devices employ various methods to store and retrieve information:

1. Primary Memory (RAM):

Primary memory, commonly referred to as RAM (Random Access Memory), stores data that the computer is actively using or processing. It's volatile memory, meaning it is temporary and erased when the computer is powered off.

2. Secondary Storage Devices:

Secondary storage devices include various non-volatile storage options that hold data even when the power is turned off. These devices provide long-term data retention and larger storage capacities compared to primary memory.

3. Audio and Video Cassettes:

Audio and video cassettes use magnetic tape to store analog audio or video signals. Data is recorded linearly on the tape and can be played back using appropriate players.

4. Magnetic Disks:

Magnetic disks, like floppy disks, store data using magnetized areas on the disk's surface. They offer random access to data and have been largely replaced by more modern storage technologies.

5. Hard Disks:

Hard disks use rapidly spinning disks coated with a magnetic material to store digital data. They offer large storage capacities and fast access times, making them common in computers.

6. Compact Disc (CDs):

CDs use a digital optical storage technique where data is encoded as tiny pits on the reflective surface of the disc. A laser reads the pits, and the data is converted into digital information.

7. Flash Drive:

Flash drives, also known as USB drives or thumb drives, utilize flash memory to store data. Flash memory is non-volatile and retains data even when the drive is disconnected from a power source. It's popular for portable and easy data storage.

WORD PROCESSING, DATA MANAGEMENT AND CONTROL

WORD PROCESSING

Word Processing involves creating, editing, and formatting text documents using software known as word processors. Users input and manipulate text, and can add images and tables. These applications offer formatting tools, fonts, and layouts for professional documents. Word processing includes spell check, grammar correction, and electronic saving/printing, enhancing written communication efficiency.

DATA MANAGEMENT – MONITORING AND CONTROL

Data management encompasses the structured handling of data for informed decision-making and efficient operations. In the context of monitoring and control, it pertains to the systematic management of data from sources like sensors and devices. This involves data acquisition, validation, transformation, and secure storage, ensuring accuracy and integrity. Effective data management empowers organizations to monitor, analyze trends, detect anomalies, and optimize processes for improved efficiency and product/service quality.

INTERNET

Internet is a global network of interconnected computers and servers that allows users to access and share information and resources. It facilitates communication, collaboration, and the exchange of data on a global scale.

Two main services that are widely used on the Internet are:

Web Browsers:

Web Browsers are software applications used to access and interact with the World Wide Web. They interpret web page code (HTML, CSS, and JavaScript) and display the content to users. Popular browsers include Chrome, Firefox, Safari, and Edge.

Electronic Mail (Email):

Electronic Mail (Email) is a method of sending and receiving digital messages and files over the internet. It enables individuals and organizations to communicate efficiently and quickly across distances. Email services provide features like message composition, attachment handling, and organization through folders or labels.

Fast Communication: Swift exchange of information and messages, ensuring quick transmission and response.

Cost-Free Services: Utilization of services without incurring financial charges or fees.

More efficient: Its refers to an improved process, system, or approach that accomplishes tasks with higher effectiveness, reduced waste of resources (such as time, energy, or materials), and optimized use of available inputs to achieve desired outcomes.